24-05-18(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
MAY-JUNE 2018

Max. Marks: 100 **Duration: 3 Hrs**
Class: SY BTechSemester: IV
Name of theCourse: Applied Mathematics-IV Branch: COMP/IT
Course Code:UCEC401/UITC401

Instructions:
(1) All Questions are Compulsory
(2) Figures to right indicate full marks. Each sub-question has equal marks.

Question NO.		Questions	Marks
Q.1	A	From the following data calculate Spearmen's rank correlation between x and y. x: 12, 17, 22, 27, 32 y: 113, 119, 117, 115, 121	05
	B	**Attempt any TWO of the following**	16
	(i)	The equation of the two lines of regression for a bivariate data are: $x+6y=6$ *and* $3x+2y=10$ Find i) mean values of x and y, ii) regression coefficient, iii) correlation coefficient.	
	(ii)	Obtain two lines of regression and coefficient of correlation from the following data- x: 62, 64, 65, 69, 70, 71, 72, 74 y: 126, 125, 139, 145, 165, 152, 180, 208	
	(iii)	Fit a second degree curve to the following data and estimate the value of y when x=80. x: 10, 20, 30, 40, 50, 60, 70 y: 20, 60, 70, 80, 90, 100, 100	
Q.2	A	A & B toss fair coin alternately. One who gets a head first, wins Rs 12. A starts. Find their expectations .	05
	B	**Attempt any TWO of the following**	16
	(i)	After correcting 50 pages of the proof of a book, the reader finds that there are on the average 2 errors per 5 pages. How many pages would one expect to find 0, 1, 2 and 3 errors in 1000 pages of first print of the book?.	
	(ii)	The probability that the marks of a student chosen at random will exceed 50 (out of 100) is 0.25. Find in two different ways, the probability that out of 100 students of this group 25 to 30 will have marks more than 50.	

(iii) The distribution of marks in a certain examination was found to be normal with 23% of the candidates scoring above 60 marks & 21% candidates scoring below 40. Find the mean & standard deviation of the distribution.

Q.3

A A machine is claimed to produce nails of mean length 5 cms & standard of 0.45 cm. A random sample of 100 nails gave 5.1 as their average length. Does the performance of the machine justify the claim? Where the level of significance is 6% — **05**

B **Attempt any TWO of the following** — **16**

(i) The sales-data of an item in six shops before & after a special promotional campaign are as under

Shops	A	B	C	D	E	F
Before campaign	53	28	31	48	50	42
After campaign	58	29	30	55	56	45

Can the campaign be judged to be a success 5% level of significance?

(ii) Sandal powder is packed into packets by a machine. A random sample of 12 packets is drawn & their weights are found to be 0.49, 0.48, 0.47, 0.48, 0.49, 0.50, 0.51, 0.49, 0.48, 0.50, 0.51, 0.48 kg. Test if the average packing can be taken as 0.5 kg. (LOS 5%)

(iii) Random samples of 220 students in a college were asked to give opinion in terms of yes or no about the winning of their college cricket team in a tournament. The following data are collected.

	Class in college		
	Ist year	IInd year	IIIrd year
Yes	43	20	37
No	23	57	40

Test whether there is any association between opinion and class in college. (LOS 5%)

Q.4 **Attempt any TWO of the following** — **16**

(i) In a railway marshalling yard, goods trains arrives at a rate of 30 trains per day. Assuming that the inter-arrival time follows an exponential distribution and the service time distribution is also exponential with an average of 36 minutes, calculate (i)the mean line length (ii) the probability that the queue size exceeds 10.

If the input of trains increases to an average of 33 per day, what will be the change in (i) and (ii) ?

	(ii)	A telephone exchange has one long distance operator. The telephone company finds that during the peack load, long distance calls arrive in a poisson fashion at an average rate of 15 per hour. The length of service on these calls is approximately exponentially distributed with mean length 5 minutes. **(a)** what is the probability that a subscriber will have to wait for his long distance call during the peak hours of the day? **(b)** if the scubscriber will wait, what is the expected waiting time in the system?	
	(iii)	At a one –man barber shop, the customers arrive following Poisson process at an average rate of 5 per hour and they are served accordingly to exponential distribution with an average service rate of 10 minutes. Assuming that only 5 seats are available for waiting customers, find the average time a custome spend in the system.	
Q.5	**A**	Solve the following LPP by Simplex method *Maximise* $z = 3x_1 + 2x_2$ *subject to* $x_1 + x_2 \leq 4$ $x_1 - x_2 \leq 2$ $x_1, x_2 \geq 0$	**05**
	B	**Attempt any TWO of the following**	
	(i)	Using the penalty (Big M) method solve the following LPP. *Minimise* $z = 2x_1 + 3x_2$ *subject to* $x_1 + x_2 \geq 5$ $x_1 + 2x_2 \geq 6$ $x_1, x_2 \geq 0$	**16**
	(ii)	Using the method of Lagrangian multipliers solve the following problem *Optimise* $z = 4x_1^2 + 2x_2^2 + x_3^2 - 4x_1x_2$ *subject to* $x_1 + x_2 + x_3 = 15$ $2x_1 - 5x_2 + 2x_3 = 20$ $x_1, x_2, x_3 \geq 0$	
	(iii)	Using Kuhn-Tucker conditions, solve the following NLPP. *Maximise* $z = 2x_1 + 3x_2 - x_1^2 - x_2^2$ *subject to* $x_1 + x_2 \leq 1$ $2x_1 + 3x_2 \leq 6$ $x_1, x_2 \geq 0$	

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End Semester Exam
April - May 2018

Max. Marks:100 | **Duration:3 hrs**
Class: SY | Semester: IV
Name of the Course: Digital Communication Network | Branch: COMPUTER
Course Code:UCEC405

Instructions:
(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		**Max. Marks**
Q 1 (a)	Define Modulation. Explain the need of modulation in detail with proper examples	10
	OR	
	A radio receiver with equivalent noise bandwidth of 10 kHz a noise figure of 20 dB. If input SNR to receiver is 40 dB, determine the output SNR. What is equivalent noise temperature if ambient temperature is 27° C	10
Q 1 (b)	What is throughput of the system both in Pure ALOHA and Slotted ALOHA, if the networks transmit 200 bit-frames on a shared channel of 200 kbps and system produces a.) 1000 frames per second b.) 500 frames per second	10
Q2 (a)	A discrete memory less source has an alphabet of five symbols with their probabilities as shown in:	10
	Symbol: M1, M2, M3, M4, M5 — **Probability**: 0.4, 0.19, 0.16, 0.15, 0.10	
	Construct a Shanon Fano code for the source and calculate code efficiency, redundancy of code. Repeat same for Huffman source coding technique.	
Q2 (b)	What is meant by multiplexing? Differentiate between FDM and TDM with proper examples.	10
	OR	
	Explain the operation of DM system. Explain various types of distortions in a DM system.	10

Symbol probabilities for Q2 (a):

Symbol	M1	M2	M3	M4	M5
Probability	0.4	0.19	0.16	0.15	0.10

Q3 (a)	What is meant by Spread Spectrum? Differentiate between FHSS and DSSS OR Write short notes on any two: i. ASK ii. PSK iii. FSK	10 10
Q3 (b)	What is line coding? What are important properties of line code? Represent the data 11010011using RZ AMI and unipolar NRZ.	10
Q4 (a)	List and explain design issues for OSI layers and differentiate between connection oriented and connectionless services	10
Q4 (b)	Explain any 3 connecting devices in detail with relevant examples OR Write a note on different types of topologies	10 10
Q5 (a)	Explain Sliding window protocol using Go back-N technique. OR Draw the HDLC frame format & explain in detail the control field used in HDLC protocol for different frame types.	10 10
Q5 (b)	Explain CSMA/CA technique in detail OR Explain in detail about IEEE 802.3	10 10